

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & -M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = \mathbb{M}_{3b}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4a}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & -M_{xxyz} & 0 & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & -M_{xxyz} \\ -M_{xxxy} & M_{xxxy} & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xyz} &= 2\mathbb{T}_{3a}^{(1)} \\
M_{xxy} &= 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} \\
M_{yzz} &= -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzz} & 0 \\ -M_{yzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & 0 & M_{xxy} \\ -M_{xyz} & 0 & M_{xxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzz} & 0 \\ M_{yzz} & 0 & 0 \\ 0 & -M_{xyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xyz} &= \mathbb{T}_{3a}^{(2)} \\
M_{xxy} &= \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzz} &= \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzz} & -M_{yzz} & 0 & 0 & M_{yzz} & 0 \\ 0 & 0 & 0 & M_{yzz} & 0 & -M_{yzz} \\ M_{xyx} & M_{xyx} & M_{xyz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzz} &= \mathbb{T}_{3a}^{(3)} \\
M_{yzz} &= \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & M_{xxy} & M_{xxz} & 0 & M_{xxy} \\ 0 & 0 & 0 & -M_{xyz} & 0 & -M_{xxy} & -M_{xxz} & 0 & -M_{xxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ M_{yzz} & -M_{yzz} & 0 & 0 & M_{yzz} & 0 & 0 & M_{yzz} & 0 \\ 0 & 0 & 0 & -M_{yzz} & 0 & M_{yzz} & -M_{yzz} & 0 & M_{yzz} \\ -\frac{M_{xxx}}{2} - \frac{M_{xyx}}{2} & \frac{M_{xyx}}{2} + \frac{M_{xyx}}{2} & 0 & 0 & M_{xxz} & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xyz} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)} \\
M_{xxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{xxz} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)} \\
M_{xyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzz} &= \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)} \\
M_{yzz} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{yzz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & 0 & -M_{yzxx} & M_{yzxz} & 0 & -M_{yzxx} \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzxz} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{yzxz} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzx} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yyyy} \\ 0 & 0 & 0 \\ 0 & M_{yyyy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyy} \\ 0 & 0 & 0 \\ 0 & -M_{yyyy} & 0 \\ 0 & 0 & 0 \\ -M_{yyyy} & 0 & 0 \end{bmatrix}$$

$$M_{yyyy} = -\mathbb{M}_{4a}^{(1)}$$